STA 244 – Vehicle Price Prediction: A Regression Model

AF Regression Project 2026

Group [XX]

07 September 2026

1 Introduction and data preparation

An online marketplace wants a tool that recommends an asking `price` (in ZAR) from a vehicle's characteristics. We build an interpretable, predictive multiple linear regression model on 2119 advertised vehicles and evaluate it out-of-sample with the **root mean square error (RMSE)**, the metric used in the competition.

```r
train_raw <- read.csv("Assignment Information and Data sets-20260907/project_traindata.txt",
                      stringsAsFactors = TRUE)
# listing_id is an identifier (not a characteristic); model_year is a perfect linear
# duplicate of vehicle_age (model_year + vehicle_age = 2026), so we keep vehicle_age only.
dat <- train_raw %>% select(-listing_id, -model_year)
```

The data contain no missing values. `price` is strongly right-skewed (mean R175,562 > median R141,500) and is bounded below by zero, so a **log transformation of the response** is a natural candidate: it should symmetrise the distribution, stabilise the variance, linearise the (curved) relationships with age and mileage, and guarantee positive fitted prices.

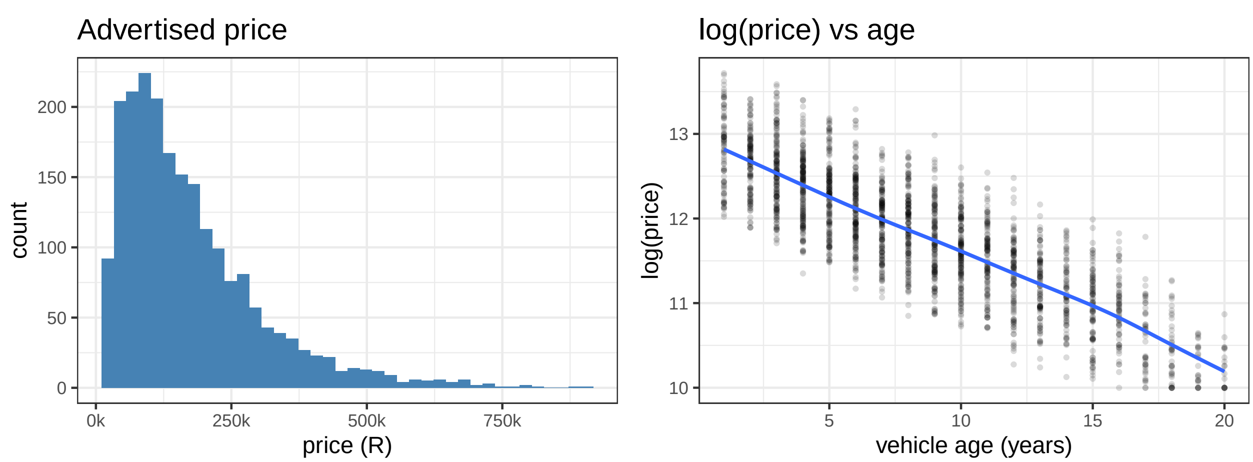

Figure 1: Left: price is right-skewed. Right: log(price) is approximately symmetric and its relationship with vehicle age is close to linear.

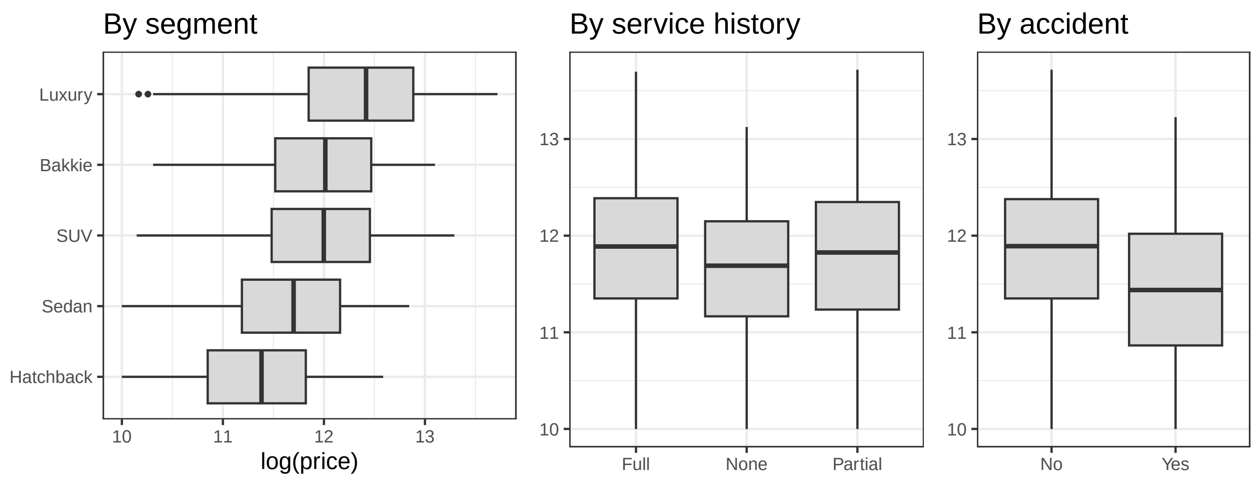

Figure 2: log(price) differs markedly across vehicle segments and worsens with poorer service history and reported accidents, confirming these are informative predictors.

2 Training and validation sets

To guard against over-fitting we randomly partition the data into an **80% training** set (used to estimate every candidate model) and a **20% validation** set (used only to compare out-of-sample prediction accuracy). An 80/20 split is standard practice: it retains enough data for stable estimation of the many factor levels while leaving a validation sample large enough for a reliable RMSE. A fixed seed makes the split reproducible.

```
set.seed(244)
idx   <- sample(seq_len(nrow(dat)), size = floor(0.80 * nrow(dat)))
train <- dat[idx, ]       # n = 1695
valid <- dat[-idx, ]      # n = 424
c(train = nrow(train), validation = nrow(valid), predictors = ncol(dat) - 1)
```

```
##      train validation predictors
##       1695        424         14
```

3 Model building

Strategy. We work on the log scale, start from the additive model with all candidate predictors, use significance testing and stepwise selection to remove uninformative variables (reducing data-collection cost), then add the non-linear and interaction terms suggested by the exploratory analysis. Each candidate is judged by validation RMSE on the original rand scale (back-transformed), supported by adjusted R^2.

Variable screening. In the full additive log-model, `drop1` F-tests show every predictor is highly significant *except* `colour` and `warranty_months` ($p \approx 0.51, 0.55$). Backward/stepwise selection by AIC agrees. We therefore drop both: the marketplace can stop collecting exterior colour and warranty length without harming predictive accuracy, saving data-collection and storage cost.

```
full_log <- lm(log(price) ~ ., data = train)
drop1(full_log, test = "F")                     # colour & warranty_months: not significant
step_aic <- step(full_log, direction = "both", trace = 0)    # confirms same reduced set
train <- train %>% select(-colour, -warranty_months)
valid <- valid %>% select(-colour, -warranty_months)
```

Non-linearity and interactions. The EDA curves for age and mileage bend slightly, so we allow second-order (quadratic) terms for both. The brief notes that a predictor's effect may differ by segment; depreciation with age is indeed steeper for luxury vehicles than bakkies, so we add a `segment` $\times$ age interaction. We compare four nested candidates:

```
f0 <- log(price) ~ .                                                      # reduced additive
f1 <- log(price) ~ . - vehicle_age + poly(vehicle_age, 2)                 # + age^2
f2 <- log(price) ~ . - vehicle_age - mileage +
        poly(vehicle_age, 2) + poly(mileage, 2)                           # + mileage^2
f3 <- log(price) ~ . - vehicle_age - mileage +
        poly(vehicle_age, 2) + poly(mileage, 2) + segment:poly(vehicle_age, 2)  # + segment x age
mods <- list(Additive = f0, `+age^2` = f1, `+mileage^2` = f2, `+seg:age` = f3)
res <- sapply(mods, function(f) {
  m  <- lm(f, data = train)
  s2 <- summary(m)$sigma^2                        # log-normal back-transform bias correction
  c(adjR2   = summary(m)$adj.r.squared,
    trRMSE  = rmse(train$price, exp(predict(m, train)) * exp(s2 / 2)),
    valRMSE = rmse(valid$price, exp(predict(m, valid)) * exp(s2 / 2)))
})
tab <- t(res)
data.frame(adjR2 = round(tab[, "adjR2"], 3),
           trainRMSE = round(tab[, "trRMSE"]),
           validRMSE = round(tab[, "valRMSE"]))
```

```
##            adjR2 trainRMSE validRMSE
## Additive   0.951     34264     35679
## +age^2     0.951     34259     35844
## +mileage^2 0.951     34214     35710
## +seg:age   0.953     33378     33822
```

The `segment`$\times$age interaction gives the best validation RMSE, and its training and validation RMSE are close, indicating the model is **not over-fitting**. This is our final model. Because predicting on the log scale and exponentiating estimates the *median* price, we apply the standard log-normal correction factor $\exp(\hat{\sigma}^2/2)$ so that fitted values target the *mean* rand price, which lowers RMSE.

```
final <- lm(f3, data = train)
sigma2 <- summary(final)$sigma^2
```

4 Model diagnostics and validation of assumptions

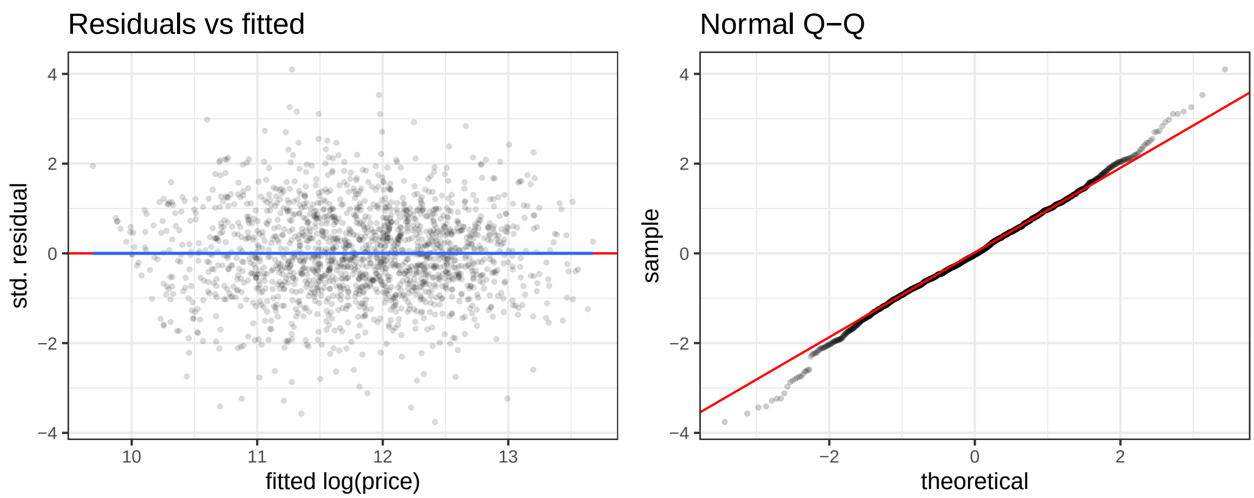

Figure 3: Residuals vs fitted show constant spread around zero with no systematic pattern (linearity + homoscedasticity); the normal Q-Q plot is close to the reference line apart from mild heavy tails.

The **linearity** and **constant-variance** assumptions are satisfied: residuals scatter randomly around zero with roughly constant spread. The Q-Q plot supports approximate **normality** with slightly heavy tails. With 1695 observations the estimators and t/F inference are robust to this mild departure (CLT). **Multicollinearity** is not a concern: generalised VIFs on the additive model are all low (below).

```
round(sort(vif(lm(log(price) ~ ., data = train))[, 3]^2), 2)  # (GVIF^(1/2df))^2 per predictor
```

```
##      seller_type         province  service_history accident_history
##             1.01             1.02             1.02             1.04
##             make        fuel_type     transmission          segment
##             1.05             1.20             1.23             1.74
##          mileage  previous_owners      vehicle_age      engine_size
##             1.95             2.24             3.09             4.39
```

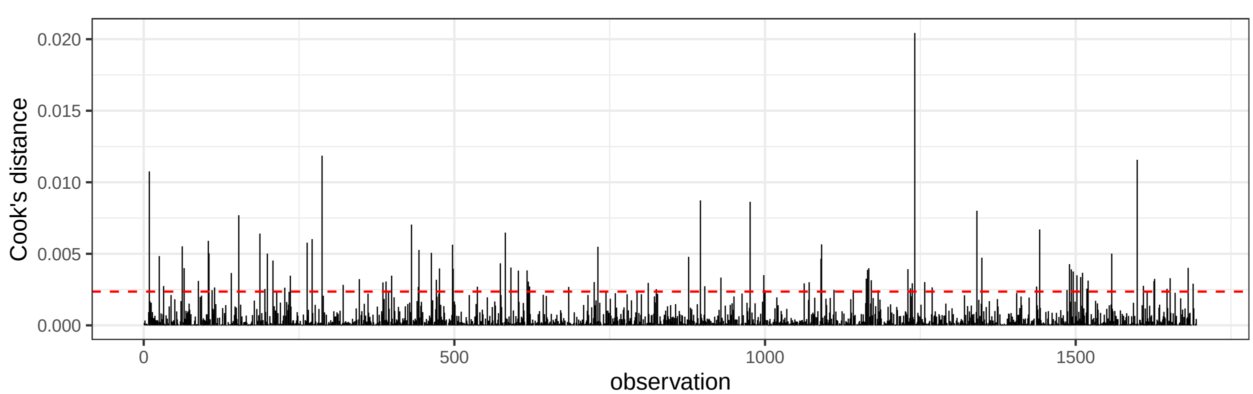

Figure 4: Cook's distance: a few advertisements are mildly influential but none exceeds the 4/n rule-of-thumb by a large margin, so no observations are removed.

We inspected the highest-leverage points (very low-mileage or very high-priced luxury listings). They are plausible genuine adverts, not data errors, so — following the brief — **no observations are removed**.

5 Interpretation and business use

Coefficients are read on the log scale, so $100(e^{\beta} - 1)\%$ is the percentage change in price for a one-unit increase in a predictor, holding others fixed.

term	pct change in price
transmissionManual	-4.3%
fuel_typePetrol	-3.0%
engine_size	+9.8%
service_historyNone	-16.3%
service_historyPartial	-5.5%
accident_historyYes	-10.7%
previous_owners	-1.9%
seller_typePrivate	-3.8%

Key drivers: **age and mileage** dominate — price falls steeply as a car gets older and is driven more (the quadratic terms capture depreciation flattening out for older vehicles). A **reported accident** and **no service history** each cut price substantially, while a **larger engine**, **automatic transmission** and **dealership** sales command a premium. Segment strongly shifts the price level, and the `segment`×age interaction shows luxury vehicles depreciate fastest.

How the business uses this. For a new advert the tool returns a recommended price $\widehat{\text{price}} = \exp(\hat{\eta})\exp(\hat{\sigma}^2/2)$. Adverts whose asking price sits far above (below) this prediction can be flagged to buyers as relatively expensive (a bargain). The screening step also tells the marketplace it can safely stop collecting `colour` and `warranty_months`.

6 Final model and competition predictions

We refit the chosen specification on the **full training data** (train + validation) to use all information for the final predictions, then predict the test set and write the competition file.

```
final_full <- lm(f3, data = dat %>% select(-colour, -warranty_months))
s2_full    <- summary(final_full)$sigma^2

test <- read.csv("Assignment Information and Data sets-20260907/project_testdata.txt",
                 stringsAsFactors = TRUE)
pred_price <- exp(predict(final_full, test)) * exp(s2_full / 2)

submission <- data.frame(listing_id = test$listing_id, price = round(pred_price))
write.csv(submission, "competition_predictions.csv", row.names = FALSE)
head(submission, 3)
```

```
##   listing_id  price
## 1    SA01243 118021
## 2    SA00409  36240
## 3    SA01848 197289
```

The validated out-of-sample RMSE of approximately R33,822 (against a median price of R141,500) indicates the model predicts advertised prices well while remaining interpretable and parsimonious.